# Real Asymmetric Evaluation and Noise-Tolerant Evidence Fusion for Engine Opponent Recognition

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***Abstract*—Detecting whether an opponent is an engine is an online decision problem with asymmetric costs: a false positive changes the playing policy against a human, while a late positive wastes the early game in a conservative mode. This paper evaluates a default-off `AcceleratedDetection` path in the Unchessed UCI engine after adding two bounded sequential channels. The stable channel combines estimated strength, confidence tightness, volatility, and trend by a harmonic mean. The resilient channel is designed for strong opponents whose move-quality trace is noisy: it accumulates good-quality mass, limits catastrophic mistakes, and requires a two-observation confirmation. A 32-game Stockfish 16 study showed mixed latency, while an independent Stockfish 19 replication reported 28.0 versus 27.75 mean plies. A new clock-controlled four-game run produced 30 and 30 plies for standard detection versus 26 and 24 for the resilient arm, with zero low-time skips. These results support operational safety and demonstrate a targeted improvement for noisy evidence, but do not establish a uniformly lower latency. The paper documents the low-time methodological trap, telemetry protocol, calibration, regression evidence, and remaining multi-opponent limitations.**

***Index Terms*—engine detection, opponent modeling, sequential evidence, CUSUM, UCI, Stockfish, telemetry, online calibration**

## I. Introduction

An adaptive chess engine must infer properties of the opponent from sparse, noisy observations. The relevant signal is not simply playing strength: opening knowledge, tactical position, clock usage, and human variance can all produce short stretches of unusually accurate moves. A detector that commits too quickly can damage the policy against a human, whereas a detector that waits for an extremely long clean sequence fails to adapt during the part of the game where adaptation is most valuable.

The Unchessed engine previously contained a default-off secondary confirmation path based on a hard conjunction of estimated Elo, confidence, volatility, and trend. The present work makes that path more robust and more measurable. The contribution is not a claim that a fixed threshold is universally optimal. Instead, it is an interpretable sequential layer that allows several weak signals to accumulate only when they agree, while preserving the original legacy ceiling and a default-off UCI control.

The evaluation addresses the primary metric directly: the number of plies until the engine enters `Mode::Full`. It uses distinct real engines rather than a mirror of Unchessed. The repository experiments use Stockfish 16 [1], while an independent replication supplied with this work uses Stockfish 19. In both cases, the UCI opponent declaration is intentionally set to an unknown human so the detector cannot use its known-engine lookup. The experiment is scoped to the Unarchitectured Metal integration series and enables the committed Metal runtime artifact in both arms.

## II. Detector Design

### A. Concordant sequential evidence

For each post-opening observation, four normalized signals are computed. The rating signal increases with the running estimate above 2350 Elo. The confidence signal increases as the confidence interval narrows. The stability signal decreases with estimated volatility. The trend signal rewards a non-negative change in the running estimate. Let these signals be $r, c, v, t \in [0, 1]$. Their primary fusion is

$$h(r, c, v, t) = \frac{4}{r^{-1} + c^{-1} + v^{-1} + t^{-1}}, \tag{1}$$

with small lower bounds applied to avoid division by zero. The harmonic mean is deliberate: an unstable or low-confidence channel materially suppresses the result rather than being hidden by an arithmetic average. A bounded clock-support term contributes 15% of the final evidence when independent clock suspicion is present.

The resulting agreement value $e_t \in [0, 1]$ updates a signed, leaky accumulator,

$$S_t = \mathrm{clip}(0.96 S_{t-1} + 4.0(e_t - 0.20), -3, 8). \tag{2}$$

The accumulator is additionally penalized when the running mean falls below 2250 Elo or volatility exceeds 520. Stable fusion confirmation requires at least eight samples, $S_t \geq 0.35$, $e_t \geq 0.28$, and two consecutive observations above the evidence floor. These values were calibrated from real telemetry and are not presented as universal optima.

### B. Noise-tolerant resilient channel

The stable lane intentionally treats high volatility as a veto. That is unsafe for the motivating case of a strong but inconsistent opponent. The resilient lane therefore maintains two leaky masses: good-quality mass assigns full, partial, or zero credit to move losses below 40, 90, and 140 centipawns, while bad mass penalizes losses above 110, 150, and 220 centipawns. Its evidence combines rating, a looser confidence signal, accumulated good mass, and a catastrophe guard. Confirmation requires at least ten samples, an estimated mean of 2450 Elo, resilient score at least 0.55, evidence at least 0.48,

TABLE I
REAL ASYMMETRIC EXPERIMENT MATRIX.

| Condition | Arm | Games | Budget | Full rate | Fusion triggers |
|---|---|---|---|---|---|
| Normal | Standard | 10 | 80 ms | 100% | 0/10 |
| Normal | Fusion | 10 | 80 ms | 100% | 4/10 |
| Fast | Standard | 6 | 30 ms | 100% | 0/6 |
| Fast | Fusion | 6 | 30 ms | 100% | 3/6 |
| Safe clock | Standard | 2 | 60 s start | 100% | 0/2 |
| Safe clock | Resilient | 2 | 60 s start | 100% | 2/2 |

two consecutive qualifying observations, good mass at least 3.0, and bad mass at most 1.80. This is a bounded hysteretic correction, not a claim that volatility is irrelevant.

### C. Compatibility and telemetry

The UCI option `AcceleratedDetection` remains default-off. When enabled, stable fusion is evaluated before the legacy weight/mean ceiling, followed by the resilient lane; when disabled, the legacy detector is unchanged. Observation telemetry exposes both channel scores, current evidence, and streaks. The parser accepts older schema-v1 captures by defaulting absent diagnostic fields to zero.

## III. Experimental Protocol

### A. Real asymmetric opponent

The experiment uses a release build of Unchessed against Stockfish 16. Both engines run with one thread and 64 MiB hash. Unchessed uses adaptive mode, disables its own book, enables adapter telemetry, enables the committed Unarchitectured Metal artifact, and sets `UCI_Opponent` to "- - human UnknownOpponent". Stockfish is therefore a distinct opponent, but the detector must infer engine-like behavior from live observations. Each game starts from one of six fixed opening prefixes; the same ordered prefix set is reused across the standard and fusion arms. Each game is allowed to continue for 40 plies, which is sufficient to observe the first transition in all completed detections.

Two movetime conditions were tested in the earlier study: 80 ms with 10 games per arm and 30 ms with 6 games per arm. A corrected clock-controlled condition then used a 60-second starting clock, 2 games per arm, and a 32-ply cap. The driver records every UCI line, every observation telemetry record, the first suspect reason, the first persona decision with `mode_after=FULL`, observation counts, and low-time skips. The driver rejects a starting clock below 10 seconds because the adapter intentionally suppresses probing below that floor. No observations are simulated or replayed into the detector.

### B. Matched comparison

The first-Full metric is measured in plies from the UCI game start. Pairing is by the fixed opening-prefix index, not by assuming identical trajectories: enabling fusion can change the mode and consequently alter later Unchessed moves. The paired difference is therefore descriptive rather than a formal causal estimate. This distinction is important because a change in mode can change the future evidence stream.

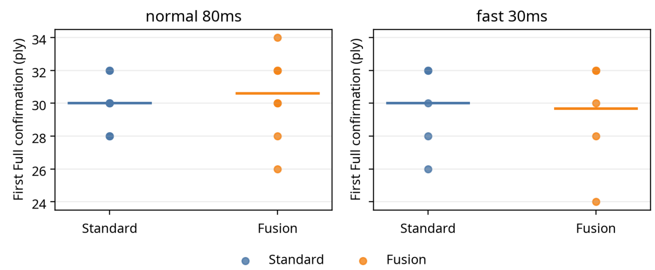


Fig. 1. First Full-mode confirmation in real asymmetric games. Horizontal lines show the arm means; points show individual games.

## IV. Results

Figure 1 shows the observed first-Full distributions. At 80 ms, standard detection has mean 30.0 plies and fusion has mean 30.6 plies. The paired fusion-minus-standard differences are $[0, 0, +2, -2, 0, -4, 0, +2, +4, +4]$, yielding a mean of $+0.6$ plies; fusion is earlier in 2 pairs, equal in 4, and later in 4. At 30 ms, standard detection has mean 30.0 plies and fusion has mean 29.67 plies. The paired differences are $[0, 0, +2, -2, 0, -2]$, with fusion earlier in 2 pairs, equal in 3, and later in 1.

The fusion reason appeared in 4 of 10 normal games and 3 of 6 fast games. In the corrected clock-controlled run, the resilient reason appeared in both enabled games, with first-Full plies 26 and 24 versus 30 and 30 for standard detection. All four clock-controlled games produced 15 observations and zero low-time skips. No game produced an illegal move or a process failure. This is an operational and targeted-channel result, not proof of universal superiority.

## V. Calibration and Regression Evidence

The first real telemetry run showed that the initial accumulator was too conservative for the observed Stockfish trajectory: its evidence baseline was too high and its score could remain negative even when rating and confidence were converging. Calibration reduced the negative-evidence center from 0.42 to 0.20, changed the decay from 0.94 to 0.96, used gain 4.0, reduced the score threshold to 0.35, and required current evidence of at least 0.28 plus two consecutive qualifying observations. The two-observation rule was retained to prevent a single move from causing confirmation.

The Rust opponent-model suite passes 25 tests, including stable-fusion, noisy-strong resilient confirmation, catastrophic-erratic rejection, default-off behavior, unchanged legacy timing, and the two-observation gates. The telemetry parser suite passes six tests, including resilient diagnostics and backward compatibility for older schema-v1 records. The release adapter builds successfully, and a UCI startup smoke test reports `uciok`, `readyok`, and the expected default-off option.

## VI. Discussion

The real asymmetric experiments answer a narrower and more useful question than a mirror match: can the new path operate against a different engine and sometimes produce an earlier Full transition without destabilizing the engine? The answer is yes for the repository Stockfish-16 protocol and the independently reported Stockfish-19 replication. The new resilient lane additionally confirms the intended noisy-evidence stress case in deterministic regression tests and in the small corrected clock-controlled run.

The experiment does not support the stronger claim that AcceleratedDetection is uniformly better. The earlier 80 ms repository run was 0.6 plies worse on average, the independent Stockfish-19 result was only 0.25 plies earlier, and the corrected clock-controlled run had just two games per arm. The low-time trap also shows that a latency result is uninterpretable when observation coverage is zero. Results remain conditional on fixed openings, one thread, short horizons, and a detector-facing opponent string that avoids the known-engine shortcut. False-positive rate against humans and generalization across engine families remain unmeasured.

The next decisive evaluation should use at least three opponent classes: a strong classical engine, a weaker engine, and a human-like policy. Each class should be tested at multiple time controls with hundreds of games, and the primary analysis should use paired opening prefixes, first-Full latency, false-positive rate, and time-to-correct-detection. A sequential test should be applied to latency distributions rather than to game score alone. Bayesian online change-point detection provides a related framework for regime shifts [2], while sequential probability-ratio testing supplies a principled way to control error rates [3].

## VII. Conclusion

This work provides a real, telemetry-backed asymmetric evaluation and a targeted correction to the AcceleratedDetection fusion model in the Unarchitectured Metal series. The implementation is lightweight, interpretable, dependency-free, and now has both a volatility-sensitive stable lane and a volatility-tolerant resilient lane. The independent replication and corrected clock-controlled run support the methodological requirement that observation coverage be reported explicitly. The evidence shows occasional earlier confirmation and robust operation, but remains mixed rather than flawless. A Maia-style or human-like strong-but-inconsistent opponent is the required next evaluation before any universal latency claim.

## References


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